We have:

$$
\begin{aligned}
{}^{\rho,\tau}g_{ij}(\theta) &= \int (\partial_i \rho(p_\theta(x)))\,(\partial_j \tau(p_\theta(x)))\,\mathrm{d}\nu(x), && (133)\\
&= \int \rho'(p_\theta(x))\tau'(p_\theta(x))\,(\partial_i p_\theta(x))\,(\partial_j p_\theta(x))\,\mathrm{d}\nu(x), && (134)\\
&= \int f''(\rho(p_\theta(x)))\,(\partial_i \rho(p_\theta(x)))\,(\partial_j \rho(p_\theta(x)))\,\mathrm{d}\nu(x), && (135)\\
&= \int (f^*)''(\tau(p_\theta(x)))\,(\partial_i \tau(p_\theta(x)))\,(\partial_j \tau(p_\theta(x)))\,\mathrm{d}\nu(x). && (136)
\end{aligned}
$$

3.13 Dually flat spaces and canonical Bregman divergences

We have described how to build a dually flat space from any strictly convex and smooth generator F: A Hessian structure is built from $F(\theta)$ with Riemannian Hessian metric $\nabla^2 F(\theta)$, and the convex conjugate $F^*(\eta)$ (obtained by the Legendre-Fenchel duality) yields the dual Hessian structure with Riemannian Hessian metric $\nabla^2 F^*(\eta)$. The dual connections ∇ and ∇^* are coupled with the metric. The connections are defined by their respective Christoffel symbols $\Gamma(\theta) = 0$ and $\Gamma^*(\eta) = 0$, showing that they are flat connections.

Conversely, it can be proved [8] that given two dually flat connections ∇ and ∇^*, we can reconstruct two dual canonical strictly convex potential functions $F(\theta)$ and $F^*(\eta)$ such that $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$. The canonical divergence A_{F,F^*} yields the dual Bregman divergences B_F and B_{F^*}.

The only symmetric Bregman divergences are squared Mahalanobis distances M_Q^2 [20] with the Mahalanobis distance defined by:

$$M_Q(\theta, \theta') = \sqrt{(\theta' - \theta)^\top Q(\theta' - \theta)}. \tag{137}$$

Let $Q = LL^\top$ be the Cholesky decomposition of a positive-definite matrix $Q \succ 0$. It is well-known that the Mahalanobis distance M_Q amounts to the Euclidean distance on affinely transformed points:

$$
\begin{aligned}
M_Q^2(\theta, \theta') &= \Delta\theta^\top Q \Delta\theta, && (138)\\
&= \Delta\theta^\top L L^\top \Delta\theta, && (139)\\
&= M_I^2(L^\top\theta, L^\top\theta') = \|L^\top\theta - L^\top\theta'\|^2, && (140)
\end{aligned}
$$

where $\Delta\theta = \theta' - \theta$.

The squared Mahalanobis distance M_Q^2 does not satisfy the triangle inequality, but the Mahalanobis distance M_Q is a metric distance. We can convert a Mahalanobis distance M_{Q_1} into another Mahalanobis distance M_{Q_2}, and vice versa, as follows:

Proof. Let us write matrix $Q = L^\top L \succ 0$ using the Cholesky decomposition. Then we have

$$M_Q(\theta_1, \theta_2) = M_I(L^\top\theta_1, L^\top\theta_2) \Leftrightarrow M_I(\theta_1, \theta_2) = M_Q((L^\top)^{-1}\theta_1, ((L^\top)^{-1}\theta_2). \tag{141}$$

Then we have for two symmetric positive-definite matrices $Q_1 = L_1^\top L_1 \succ 0$ and $Q_2 = L_2^\top L_2 \succ 0$:

$$M_{Q_1}(\theta_1, \theta_2) = M_I(L_1^\top\theta_1, L_1^\top\theta_2) = M_{Q_2}((L_2^\top)^{-1}L_1^\top\theta_1, (L_2^\top)^{-1}L_1^\top\theta_2). \tag{142}$$

It follows that we have:

$$M_{Q_1}(\theta_1, \theta_2) = M_{Q_2}((L_2^\top)^{-1}L_1^\top\theta_1, (L_2^\top)^{-1}L_1^\top\theta_2). \tag{143}$$

□

We have $M_Q^2(\theta_1, \theta_2) = B_F(\theta_1, \theta_2)$ (Bregman divergence) with $F(\theta) = \frac{1}{2}\theta^\top Q\theta$ for a positive-definite matrix $Q \succ 0$. The convex conjugate $F^*(\eta) = \frac{1}{2}\eta^\top Q^{-1}\eta$ (with $Q^{-1} \succ 0$). We have $\eta = Q^{-1}\theta$ and $\eta = Q\theta$. We have the following identity between the *dual Mahalanobis divergences* M_Q^2 and $M_{Q^{-1}}^2$:

$$M_Q^2(\theta_1, \theta_2) = M_{Q^{-1}}^2(\eta_1, \eta_2). \tag{144}$$

When the Bregman generator is based on an integral, i.e., the log-normalizer $F(\theta) = \log\left(\int \exp(\langle t(x), \theta\rangle \mathrm{d}\mu(x)\right)$ for exponential families $\mathcal{E}$, or the negative Shannon entropy $F(\theta) = \int m_\theta(x) \log m(\eta) \mathrm{d}\mu(x)$ for mixture families $\mathcal{M}$, the associated Bregman divergences $B_{F,\mathcal{E}}$ or $B_{F,\mathcal{M}}$ can be relaxed and interpreted as a statistical distance. We explain how to obtain the reconstruction below:

- Consider an exponential family $\mathcal{E}$ of order D with densities defined according to a dominating measure μ:

$$\mathcal{E} = \{p_\theta(x) = \exp(\theta^\top t(x) - F(\theta)) \ : \ \theta \in \Theta\}, \tag{145}$$

where the natural parameter θ and the sufficient statistic vector $t(x)$ belong to $\mathbb{R}^D$. We have the integral-based Bregman generator:

$$F(\theta) = F_{\mathcal{E}}(p_\theta) = \log\left(\int \exp(\theta^\top t(x)) \mathrm{d}\mu(x)\right), \tag{146}$$

and the dual convex conjugate

$$F^*(\eta) = -h(p_\theta) = \int p(x) \log p(x) \mathrm{d}\mu(x), \tag{147}$$

where $h(p) = -\int p(x) \log p(x) \mathrm{d}\mu(x)$ denotes Shannon's entropy.

Let $\lambda(i)$ denotes the i-th coordinates of vector λ, and let us calculate the *inner product* $\theta_1^\top \eta_2 = \sum_i \theta_1(i)\eta_2(i)$ of the Legendre-Fenchel divergence. We have $\eta_2(i) = E_{p_{\theta_2}}[t_i(x)]$. Using the linear property of the expectation $E[\cdot]$, we find that $\sum_i \theta_1(i)\eta_2(i) = E_{p_{\theta_2}}\left[\sum_i \theta_1(i) t_i(x)\right]$. Moreover, we have $\sum_i \theta_1(i) t_i(x) = (\log p_{\theta_1}(x)) + F(\theta_1)$. Thus we have:

$$\theta_1^\top \eta_2 = E_{p_{\theta_2}}\left[\log p_{\theta_1} + F(\theta_1)\right] = F(\theta_1) + E_{p_{\theta_2}}\left[\log p_{\theta_1}\right]. \tag{148}$$

It follows that we get

$$\begin{aligned} B_{F,\mathcal{E}}[p_{\theta_1} : p_{\theta_2}] &= F(\theta_1) + F^*(\eta_2) - \theta_1^\top \eta_2, & (149)\\ &= F(\theta_1) - h(p_{\theta_2}) - E_{p_{\theta_2}}[\log p_{\theta_1}] - F(\theta_1), & (150)\\ &= E_{p_{\theta_2}}\left[\log \frac{p_{\theta_2}}{p_{\theta_1}}\right] =: D_{\mathrm{KL}^*}[p_{\theta_1} : p_{\theta_2}], & (151) \end{aligned}$$

By relaxing the exponential family densities p_{θ_1} and p_{θ_2} to be arbitrary densities p_1 and p_2, we obtain the *reverse KL divergence* between p_1 and p_2 from the dually flat structure induced by the integral-based log-normalizer of an exponential family:

$$\begin{aligned} D_{\mathrm{KL}^*}[p_1 : p_2] &= E_{p_2}\left[\log \frac{p_2}{p_1}\right] = \int p_2(x) \log \frac{p_2(x)}{p_1(x)} \mathrm{d}\mu(x), & (152)\\ &= D_{\mathrm{KL}}[p_2 : p_1]. & (153) \end{aligned}$$

Thus we have recovered the reverse Kullback-Leibler divergence D_{KL^*} from $B_{F,\mathcal{E}}$.

The dual divergence $D^*[p_1 : p_2] := D[p_2 : p_1]$ is obtained by swapping the distribution parameter orders. We have:

$$D^*_{\mathrm{KL}^*}[p_1 : p_2] := D_{\mathrm{KL}^*}[p_2 : p_1] = E_{p_1}\left[\log \frac{p_1}{p_2}\right] =: D_{\mathrm{KL}}[p_1 : p_2], \tag{154}$$

and $D_{\mathrm{KL}^*}[p_1 : p_2] = D^*_{\mathrm{KL}^*}[p_2 : p_1] = D_{\mathrm{KL}}[p_2 : p_1]$.

To summarize, the canonical Legendre-Fenchel divergence associated with the log-normalizer of an exponential family amounts to the statistical reverse Kullback-Leibler divergence between p_{θ_1} and p_{θ_1}

(or the KL divergence between the swapped corresponding densities): $D_{\mathrm{KL}}[p_{\theta_1} : p_{\theta_2}] = B_F(\theta_2 : \theta_1) = A_{F,F^*}(\theta_2 : \eta_1)$. Notice that it is easy to check that $D_{\mathrm{KL}}[p_{\theta_1} : p_{\theta_2}] = B_F(\theta_2 : \theta_1)$ [14, 16]. Here, we took the opposite direction by constructing D_{KL} from B_F.

We may consider an auxiliary carrier term $k(x)$ so that the densities write $p_\theta(x) = \exp(\theta^\top t(x) - F(\theta) + k(x))$. Then the dual convex conjugate writes [88] as $F^*(\eta) = -h(p_\theta) + E_{p_\theta}[k(x)]$.

Notice that since the Bregman generator is defined up to an affine term, we may consider the equivalent generator $F(\theta) = -\log p_\theta(\omega)$ instead of the integral-based generator. This approach yields ways to build formula bypassing the explicit use of the log-normalizer for calculating various statistical distances [94].

- In this second example, we consider a mixture family

$$\mathcal{M} = \left\{ m_\theta = \sum_{i=1}^{D} \theta_i p_i(x) + (1 - \sum_{i=1}^{D} \theta_i) p_0(x) \right\}, \tag{155}$$

where $p_0, \ldots, p_D$ are $D+1$ linearly independent probability densities. The integral-based Bregman generator F is chosen as Shannon negentropy:

$$F(\theta) = F_{\mathcal{M}}(m_\theta) = -h(m_\theta) = \int m_\theta(x) \log m_\theta(x) \mathrm{d}\mu(x). \tag{156}$$

We have

$$\eta_i = [\nabla F(\theta)]_i = \int (p_i(x) - p_0(x)) \log m_\theta(x) \mathrm{d}\mu(x), \tag{157}$$

and the dual convex potential function is

$$F^*(\eta) = -\int p_0(x) \log m_\theta(x) \mathrm{d}\mu(x) = h^\times(p_0 : m_\theta), \tag{158}$$

i.e., the cross-entropy between the density p_0 and the mixture m_θ. Let us calculate the inner product $\theta_1^\top \eta_2$ of the Legendre-Fenchel divergence as follows:

$$\begin{aligned} \sum_i \theta_1(i) \int (p_i(x) - p_0(x)) \log m_{\theta_2}(x) \mathrm{d}\mu(x) &= \int \sum_i \theta_1(i) p_i(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x) \\ &\quad - \sum_i \theta_1(i) p_0(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x). \end{aligned} \tag{159}$$

That is

$$\theta_1^\top \eta_2 = \int \sum_i \theta_1(i) p_i \log m_{\theta_2} \mathrm{d}\mu - \sum_i \theta_1(i) p_0 \log m_{\theta_2} \mathrm{d}\mu. \tag{160}$$